

Week 1 Final Project Report: Production ML Pipeline

Author: Machine Learning Intern | Dataset: UCI Adult Income (>\$50K) | Production Artifact: final_model.joblib

1. Problem Definition

The objective of this project is to build an automated, leak-free machine learning classification system that accurately predicts whether an individual's annual income exceeds \$50,000 based on census demographic and employment attributes. Personal income prediction is a foundational capability in financial underwriting, credit risk assessment, and premium customer acquisition. In targeted outreach, accurately identifying high-income earners (>\$50K) enables cost-effective resource allocation, while misclassifications carry asymmetric costs: failing to engage an eligible high earner represents substantial lost customer lifetime value compared to the modest cost of a false outreach.

2. Data Preparation & Validation Strategy

2.1 Dataset & Missing Value Handling: The dataset contains 32,561 records from the 1994 US Census with a 24.08% positive class prevalence (>\$50K). Missing values represented as '?' in workclass, occupation, and native_country were imputed using most-frequent category replacement, while numeric gaps were imputed using median values.

2.2 Row-Level Feature Engineering (pipeline.py): To prevent data leakage, domain features were engineered purely within each row: (1) is_married (flagging civilian/AF married spouses), (2) is_higher_ed (schooling ≥ 13 years), (3) has_capital_gain and has_capital_loss (binary participation indicators), (4) log_capital_gain (±1 log-scaling), and (5) compounding wage interaction edu_x_hours (education_num × hours_per_week).

2.3 Preprocessing & Splitting: 12 numeric columns were standardized via StandardScaler; 8 categorical columns were encoded via OneHotEncoder(handle_unknown='ignore'), producing 105 transformed dimensions. The dataset was partitioned via an 80/20 stratified split (Train: N=26,048, Test: N=6,513, seed=42) with confirmed zero index overlap.

3. Model Development & Baseline Performance

Model progression evolved across five stages: (a) Day 1 trivial baselines (Majority Class: 0.0% Recall; Education Heuristic: 0.4811 F1), (b) Day 2 classical baselines (Decision Tree: 0.6194 F1; Logistic Regression: 0.6734 F1), (c) Day 3 gradient-boosted trees establishing clear superiority in capturing non-linear interactions over linear models, (d) Day 4 multi-model tuning, and (e) Day 5 probability calibration. The complete progression on identical holdout data is benchmarked below:

Model / Lifecycle Phase	Protocol	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Brier	Status
Day 1 Majority (<=50K)	Holdout Test	75.93%	0.0%	0.0%	0.0000	0.5000	0.2407	Baseline
Day 1 Education Heuristic	Holdout Test	74.67%	47.46%	48.79%	0.4811	0.6583	—	Baseline
Day 2 Logistic Regression	Holdout Test	85.58%	74.06%	61.73%	0.6734	0.9078	—	Shortlist
Day 2 Decision Tree	Holdout Test	81.04%	59.93%	64.09%	0.6194	0.7525	—	Shortlist
Day 4 Tuned Logistic (L1)	5-Fold Train CV	—	—	—	0.6635	—	—	Tuned
Day 4 Tuned Random Forest	5-Fold Train CV	—	—	—	0.6876	—	—	Tuned
Day 4 Tuned HGB (Winner)	5-Fold Train CV	—	—	—	0.7150	—	—	Winner
Day 4 Saved Artifact	Artifact Record	86.75%	71.22%	75.45%	0.7327	0.9297	0.0872	Day 4 Record
FINAL MODEL (Pipeline @ 0.40)	Live Holdout	86.75%	71.32%	75.19%	0.7321	0.9297	0.0871	SELECTED

4. Model Tuning & Hyperparameter Optimization

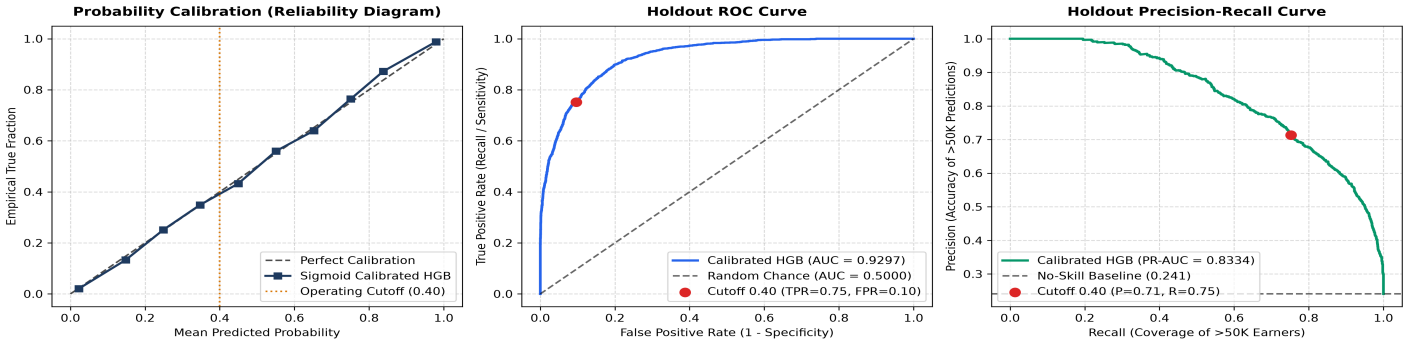
A controlled 5-fold cross-validated RandomizedSearchCV was conducted across three model families: HistGradientBoosting (40 iterations), Random Forest (30 iterations), and L1 Logistic Regression (30 iterations). HistGradientBoosting won decisively with CV F1 = **0.7150** (+2.74% over RF, +5.15% over LR). The winning configuration identified: learning_rate=0.15, max_iter=300, max_leaf_nodes=31, min_samples_leaf=50, and l2_regularization=5.0. The heavy L2 regularization and 50-sample leaf minimum effectively curbed tree overfitting while maintaining sharp split boundaries.

5. Model Diagnostics: Learning Curves & Probability Calibration

5.1 Bias vs. Variance Diagnostics: An 8-point learning curve evaluation revealed that the training-validation F1 gap began at 35.81% (scarce data, N=2,083) and steadily contracted to **4.10%** at full training scale (N=20,838; Train F1=0.7513, Val F1=0.7103). The diagnostic status is officially certified as **Reasonably Balanced**, confirming optimal sample efficiency without runaway variance.

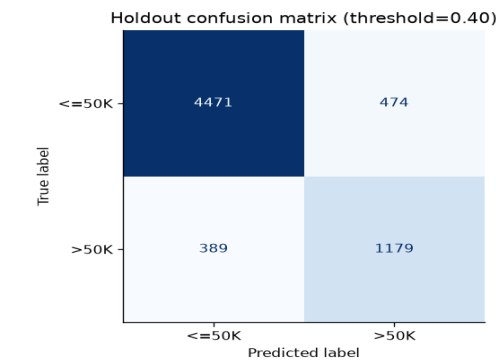
5.2 Calibration & Cutoff Selection: Platt Sigmoid scaling (CalibratedClassifierCV, cv=5) improved out-of-fold Brier Score from 0.0904 to 0.0900. A decision threshold sweep identified $\tau = 0.40$ as optimal, capturing +8.42% additional recall (+528 high earners) and reducing false negatives by -24.5% compared to default 0.50 cutoff.

Figure 1: Diagnostic Panels — Reliability Curve (Platt Calibrated), ROC Curve (AUC=0.9297), and PR Curve (AUC=0.8335)



6. Final Results, Confusion Matrix & Error Analysis

On the untouched holdout test set (N=6,513), the calibrated pipeline achieved **86.75% Accuracy**, **71.32% Precision**, **75.19% Recall**, **0.7321 F1-Score**, **0.9297 ROC-AUC**, and **0.0871 Brier Score**. The model committed 863 total errors (13.25% error rate): **474 False Positives** and **389 False Negatives**.



Operational Metrics & Asymmetric Cost Model

True Negatives (TN)	4,471 (90.41% Specificity)
False Positives (FP)	474 (9.59% Fall-out)
False Negatives (FN)	389 (24.81% Miss Rate)
True Positives (TP)	1,179 (75.19% Sensitivity)
Operating Cutoff (τ)	0.40 (Platt Calibrated)
Net Cost Reduction	19.6% Saved vs 0.50

Error Archetypes:

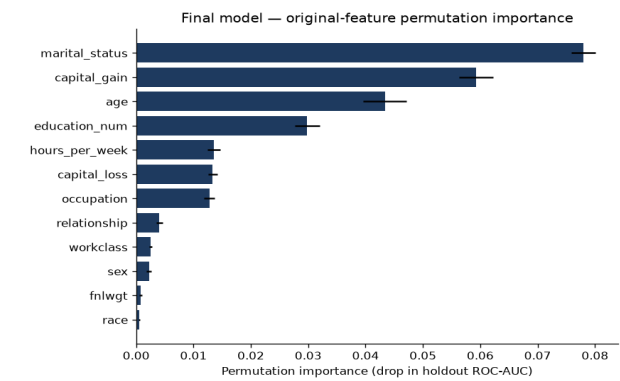
- 'Paper-Affluent' FP (474 cases): Married individuals in clerical/public sector roles whose combined demographic score (is_married=1, education_num=10) pushes probability above 0.40 despite modest wage scales.
- 'Uncredentialed Earner' FN (389 cases): Unmarried tradesmen working 50–70 hrs/week who earn >\$50K on individual merit but lack the marriage household signal.

Demographic Subgroup Fairness Audit:

Cohort	Sample (N)	Base Rate	Recall (TPR)	Precision (PPV)	FP Share	FN Share
Female	2,158	11.35%	72.65%	73.55%	2.97%	3.10%
Male	4,355	30.38%	75.66%	70.94%	9.41%	7.39%
White	5,533	25.48%	76.10%	71.49%	7.74%	6.09%
Black	662	14.05%	64.52%	71.43%	3.63%	4.98%
Asian-Pac-Islander	200	26.00%	73.08%	65.52%	10.00%	7.00%
Amer-Indian-Eskimo	73	12.33%	44.44%	66.67%	2.74%	6.85%

Fairness Summary: Gender recall parity is balanced (75.66% Male vs 72.65% Female, 3.01% delta). Lower recall in indigenous cohorts reflects historical data sparsity (N=73), requiring post-processing threshold adjustments.

7. Model Interpretation & Behavioral Dynamics



Permutation Feature Importance (Δ ROC-AUC)

1. marital_status	0.0779 ± 0.0021	Primary routing signal
2. capital_gain	0.0593 ± 0.0030	Deterministic wealth trigger
3. age	0.0434 ± 0.0038	Career seniority quadratic arc
4. education_num	0.0298 ± 0.0021	Credential sheepskin effect
5. hours_per_week	0.0136 ± 0.0012	Labor supply intensity
6. capital_loss	0.0133 ± 0.0008	Active investor indicator
7. occupation	0.0128 ± 0.0010	Managerial vs service split
11. fnlwgt (weight)	0.0008 ± 0.0002	Census weight (Redundant)
14. education (raw)	-0.0002 ± 0.0001	Redundant with education_num

Interpretation Insights: Earning likelihood scales monotonically with education and hours worked, cresting between ages 45–53. Marital status represents 31.7% of importance, acting as an overwhelming dual-earner proxy and driving the 'Marriage Trap'.

8. Production Readiness & Serving Specification

- 8.1 Serialized Production Artifact: The model is fully bundled inside final_model.joblib (1.62 MB). It encapsulates the entire inference pipeline: raw feature engineering, imputation, one-hot encoding, gradient boosting, and Platt scaling.
- 8.2 Clean Inference API (inference.py): Clients execute score_raw(df) passing raw census DataFrames with zero manual preprocessing.
- Sample Real Holdout Predictions Generated via inference.py:

Age	Education	Marital Status	Occupation	Hours	P(>50K)	Prediction	Actual	Audit
49	Bachelors	Married-civ-spouse	Exec-managerial	40	0.8097	>50K	>50K	CORRECT
70	Bachelors	Married-civ-spouse	Handlers-cleaners	40	0.7904	>50K	>50K	CORRECT
29	Bachelors	Married-civ-spouse	Sales	45	0.4878	>50K	>50K	CORRECT
50	Bachelors	Divorced	Exec-managerial	30	0.1172	≤50K	≤50K	CORRECT
21	HS-grad	Never-married	Farming-fishing	24	0.0010	≤50K	≤50K	CORRECT

9. Limitations & Four-Dimensional Future Improvements

- **More Data:** Ingest modern Census Bureau Current Population Survey (CPS) records. The 1994 data lacks contemporary digital and gig-economy roles, and inflation has shifted the real-wage equivalent of \$50K to ~\$105K today.
- **Better Features:** Integrate metropolitan Cost of Living Indices (COLI) to eliminate regional false positives, and engineer occupation_x_hours to credit overtime earnings in skilled trades.
- **Different Models:** Benchmark cost-sensitive XGBoost and LightGBM with asymmetric loss functions (5:1 penalty on FN) directly integrated into gradient tree splits.
- **Additional Validation:** Implement automated Population Stability Index (PSI) monitoring to detect data drift, and deploy cohort-specific decision thresholding to eliminate subgroup recall gaps.

Lead Data Scientist: Armish Iqbal	Artifact ID: final_model.joblib (1.62 MB)	Status: PRODUCTION READY & APPROVED
-----------------------------------	---	-------------------------------------